

PART 3 – BESS Makes Technical Sense. Is the Open Access Framework Ready?

Maharashtra has clearly placed energy storage at the centre of its renewable-energy transition. The policy direction is clear. The more difficult question for C&I Open Access projects is how storage will be implemented, metered, scheduled and commercially settled.

For developers and industrial consumers, the implementation discussion can be grouped into four fundamental questions.

1. How should stored renewable energy be metered, accounted for and commercially settled under Open Access?

Open Access settlement has traditionally been structured around generation → network → consumption. BESS introduces an additional stage: generation → storage → discharge → consumption.

This creates a connected set of questions around charging energy, round-trip losses, banking, Time-of-Day (ToD) settlement, scheduling and the treatment of energy after discharge. The same issue extends to the applicability of Transmission Charges, Wheeling Charges, Cross Subsidy Surcharge, Additional Surcharge, network losses and other system charges.

Where renewable energy used to charge a BESS has already been subjected to the applicable Open Access framework, clarity is required on how the same energy should be treated when it is subsequently discharged. The framework should avoid unintended double charging, while ensuring that legitimate network usage is appropriately recovered.

This becomes particularly important under Captive and Group Captive structures where BESS may be under different commercial structure as compared to the underlying RE asset. When renewable electricity is first charged into a battery and later consumed by captive users, a clear methodology is required for recognising such energy—and the corresponding storage losses—for captive-consumption purposes.

Further for a BESS system installed at consumer end, clear clarity is required on the contract demand implications, as additional power will be required during charging process, which will require increase in contract demand. Further the technical feasibility of the transformer and the distribution line supplying electricity to the consumer will also become important here.

A standardised energy-accounting framework is therefore essential to ensure predictable project economics and uniform implementation across stakeholders.

2. Should BESS be tied to an individual generating project, or should deployment be flexible?

Battery Energy Storage Systems can be deployed at the renewable generation site, a common Pooling Substation (PSS) or at the consumer premises. Each configuration offers different technical, operational and commercial advantages.

Project-level storage may simplify energy accounting, while centralised storage at a Pooling Substation can potentially improve utilisation across multiple renewable assets. Behind-the-Meter storage at the consumer end may provide additional value through ToD optimisation, peak-demand management, DG avoidance and improved energy resilience.

This also raises the question of whether every renewable project should require its own dedicated battery, or whether a common BESS may be shared across multiple Wind and Solar projects or multiple consumers connected through common infrastructure.

A more flexible framework could improve battery utilisation, optimise capital investment and reduce overall system cost, while continuing to meet the underlying objective of improving renewable-energy utilisation and grid reliability.

Operational flexibility is equally important. A commercially efficient BESS should ideally be capable of responding dynamically to renewable surplus, customer demand, ToD tariffs, grid conditions and, where permitted, evolving market opportunities.

3. Can a uniform storage requirement work across Solar, Wind and Hybrid projects — and how should existing assets be treated?

Solar, Wind and Wind-Solar Hybrid projects have fundamentally different generation profiles. Solar generation follows a relatively predictable daytime pattern, providing regular charging opportunities during the day and potential discharge during evening peak periods.

Wind generation, by contrast, is intermittent and may occur throughout both the day and night. This can result in different battery utilisation, charging and discharging cycles, and therefore materially different commercial outcomes compared with Solar projects.

The implementation framework should therefore consider whether a common storage requirement can achieve the intended policy objective equally across all renewable technologies, or whether the treatment should reflect the operating characteristics of the underlying generation source and the actual utilisation of storage.

A related issue is the treatment of existing renewable assets. Many industrial consumers and developers already operate Wind and Solar projects under Open Access and Captive structures. Allowing existing renewable infrastructure to be integrated with future BESS investments could improve battery utilisation, reduce incremental capital cost and accelerate storage deployment, while maintaining the intent of the policy.

4. Can BESS move beyond compliance and become a strategic energy-optimisation asset?

The long-term value of BESS extends well beyond meeting a mandatory storage requirement.

A well-designed BESS can support Time-of-Day optimisation, peak-demand management, improved renewable-energy utilisation, DG reduction, greater energy resilience and, subject to the applicable framework, participation in evolving electricity markets.

For this value to be realised, the regulatory and operational framework needs to enable the battery to capture multiple legitimate value streams rather than restricting it to a single compliance function.

Clear rules on metering architecture, Special Energy Meters, scheduling, forecasting, DSM, state-of-charge treatment, renewable versus grid charging and coordination with the SLDC / distribution licensee will therefore be important.

These are not merely technical implementation details. They directly affect battery sizing, utilisation, energy accounting, landed power cost and project bankability.

Technology enables BESS. Commercial optimisation determines its value. Regulatory clarity determines how effectively it can scale.

The next step should be a consistent implementation framework covering:

**Metering & Energy Accounting → Open Access & Captive Treatment → Deployment & Sharing
→ Scheduling & Commercial Optimisation**

Once these rules are clear, BESS can move from being viewed primarily as a policy or compliance requirement to becoming a genuine commercial optimisation asset for Maharashtra's C&I consumers.

*#BESS #BatteryEnergyStorage #MaharashtraOpenAccess #MERC #OpenAccess #GroupCaptive
#EnergyStorage #RenewableEnergy #EnergyPolicy #IntegrumEnergy*